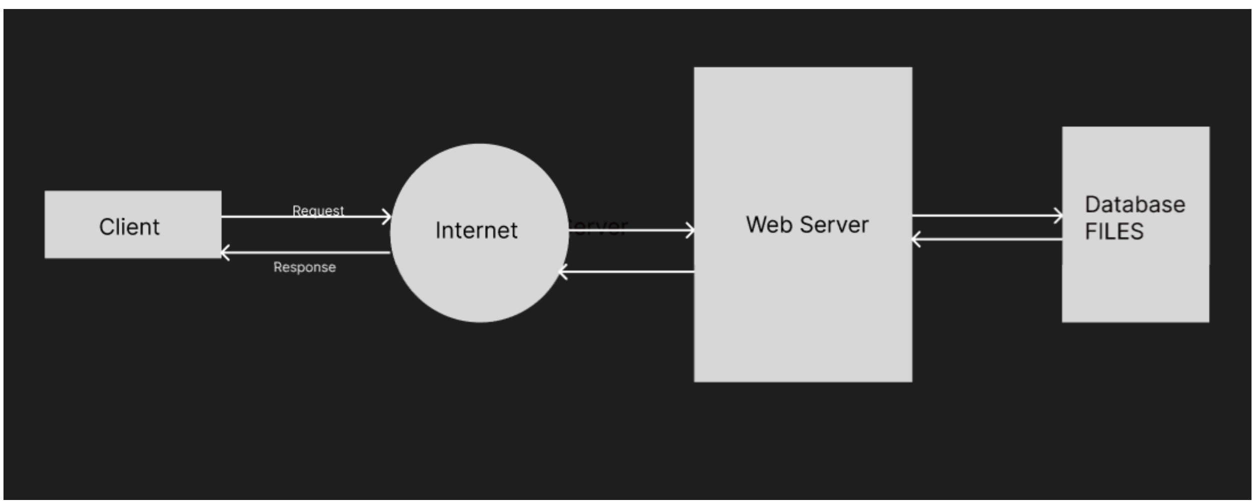
Q1 : Explain the difference between frontend, backend, and full-stack development with suitable real-world examples.
Ans :
** Frontend -> In frontend we deals with the part of a website or web app users see and interact with.
-> Here we use the technologies : HTML, CSS, JavaScript, AND React for user interaction.
-> When we open any website, everything we see buttons, product layout, and navigation is frontend.

** Backend -> IN backend we handle the server-side logic, database operations, and APIs.
-> Here we use the technologies : Node.js, Java, Python (Django), PHP, SQL
-> When we log in, our credentials are verified on a server — that's backend work.

** Full-Stack Development -> It involves both frontend and backend development. A full-stack developer can build an entire web app from end to end.
-> Here we use both frontend and backend technologies MERN Stack (MongoDB, Express, React, Node.js)

Q2 : Create a simple diagram showing how the client-server model works in web architecture.



Q3 : Describe how a browser requests and displays a web page from a web server.
Ans : **Step-by-step process:**

1. **User enters a URL** (e.g., `www.example.com`) in the browser.
2. **DNS (Domain Name System)** converts the domain name to an IP address.
3. **Browser sends an HTTP/HTTPS request** to the web server.

4. **Server processes the request**, fetches data (if needed), and sends a response (HTML, CSS, JS).
5. **Browser receives the response** and uses its **rendering engine** to display the web page visually.

Simplified example: When you open `www.youtube.com`, your browser requests the page → the server returns HTML → your browser loads the video thumbnails, layout, and JavaScript code for functionality.

Q4 : Identify and list the tools required to set up a web development environment. Explain the purpose of each.
Ans : To start web development, we need some important tools. Each tool helps us in a different way:

1. **VS Code (Visual Studio Code):**
   It is a text editor used to write and edit code. It supports many programming languages like HTML, CSS, and JavaScript.

2. **Web Browser (like Google Chrome or Firefox):**
   Used to open and test websites to see how they look and work.

3. **Node.js:**
   Allows JavaScript to run on the server. It is also used to install packages using npm (Node Package Manager).

4. **Git and GitHub:**
   Git helps to save different versions of your code, and GitHub allows you to store and share your code online with others.

5. **Live Server Extension (in VS Code):**
   It helps you to see the changes you make in your code instantly in the browser without reloading manually.

6. **Command Prompt or Terminal:**
   Used to run commands like starting a server or installing tools.

Q5 : Explain what a web server is and give examples of commonly used servers.
Ans : A **web server** is software that **stores, processes, and delivers web pages** to users upon request.

Commonly Used Web Servers:

- **Apache HTTP Server** → Open-source, widely used.

- **Nginx** → Fast, lightweight, and handles multiple requests efficiently.
- **Microsoft IIS** → Built for Windows environments.
- **Node.js** → JavaScript-based runtime for backend servers.

Example:
When you open a website, Apache or Nginx delivers the HTML file from the server to your browser.

Q6 : Define the roles of a frontend developer, backend developer, and database administrator in a project.

Ans : In a web development project, different people handle different parts of the work.

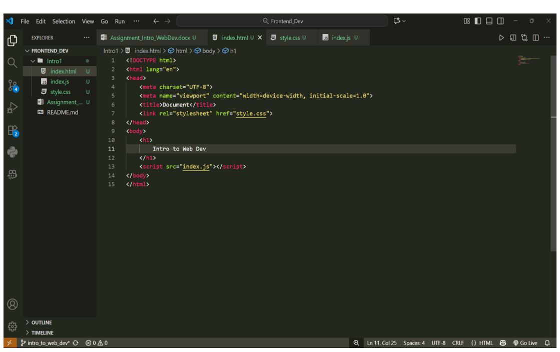
A **frontend developer** is responsible for designing how a website looks and feels. They work on the part that users see and interact with. They use technologies like HTML, CSS, and JavaScript to create buttons, forms, menus, and overall layout. Their main goal is to make the website attractive and easy to use on all devices.

A **backend developer** works on the behind-the-scenes part of a website. They handle the server, application logic, and communication with the database. They make sure that when a user performs an action (like logging in or buying a product), the server processes that request correctly. They usually use languages like Java, Python, PHP, or Node.js.

A **database administrator (DBA)** manages the database of the project. They make sure that all the data, such as user details or product information, is stored safely and can be retrieved quickly. They also take care of database security, backups, and performance.

Q7 : Install VS Code and configure it for HTML, CSS, and JavaScript development. Take a screenshot of the setup.

Ans :



Q8 : Explain the difference between static and dynamic websites. Provide an example of each.

Ans : **Difference between Static and Dynamic Websites**

A **static website** shows the same content to every visitor. The pages are created using only HTML and CSS, and the content does not change unless the developer manually edits the code. These websites are fast and simple but cannot show personalized or changing information. For example, a personal portfolio or a company's information website is usually static.

A **dynamic website**, on the other hand, displays content that can change depending on the user or data. It uses both frontend and backend technologies. Dynamic websites often take information from a database and update automatically. For example, Facebook, YouTube, and Amazon are dynamic websites because their content changes based on user actions and stored data.

Q9 : Research and list five web browsers. Explain how rendering engines differ between them.

Ans : **Web Browsers and Rendering Engines**

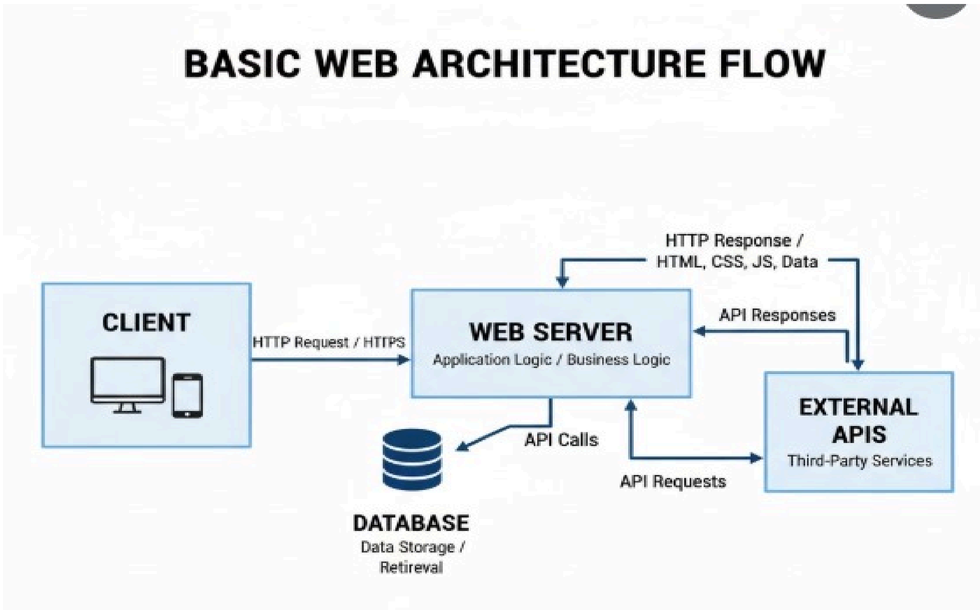
A **web browser** is a software used to open and view websites. Different browsers use different rendering engines to display web pages on the screen.

Some common web browsers are **Google Chrome**, **Mozilla Firefox**, **Safari**, **Microsoft Edge**, and **Opera**.

Each browser has its own rendering engine that converts the website's HTML, CSS, and JavaScript code into a visual web page. For example, **Google Chrome**, **Edge**, and **Opera** use the **Blink** engine; **Firefox** uses **Gecko**; and **Safari** uses **WebKit**. These engines may show small design or performance differences because each handles code slightly differently.

Q10 : Draw a labeled diagram showing the basic web architecture flow — client, server, database, and APIs.

Ans :


BASIC WEB ARCHITECTURE FLOW
CLIENT
HTTP Request / HTTPS
WEB SERVER
Application Logic / Business Logic
HTTP Response /
HTML, CSS, JS, Data
API Responses
EXTERNAL
APIS
Third-Party Services
API Calls
API Requests
DATABASE
Data Storage /
Retireval